

Project Proposal (Proposed Solution)

Date: 28 June 2026 **Team ID:** _____ **Project Name:** Rising Waters – AI Flood Prediction System

Maximum Marks: 5 Marks

Project Overview

Section	Description
Objective:	Develop an AI-powered flood prediction system that analyzes environmental factors using Machine Learning to predict flood risk and provide early warnings and safety recommendations.
Scope:	The project focuses on predicting flood risk based on environmental parameters, displaying flood probability, and helping users prepare for potential flood situations through a simple web application.

Problem Statement

Section	Description
Description	Floods cause significant loss of life and property every year. Existing warning systems are often delayed or lack location-specific predictions. The project aims to provide accurate early flood prediction using Machine Learning.
Impact	Improves disaster preparedness, supports timely decision-making, and helps reduce the impact of floods on people and property.

Proposed Solution

Section	Details
Approach	Develop a Flask-based web application integrated with an XGBoost Machine Learning model to predict flood risk from user inputs.
Key Features	AI-based flood prediction, flood risk percentage, safety recommendations, prediction history, simple user interface.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Development Laptop	Intel Core i5/i7 or Ryzen 5/7, 8 GB RAM or above
Memory	RAM	Minimum 8 GB
Storage	SSD	512 GB SSD
Software	Development Tools	Python, Flask, VS Code, HTML, CSS, JavaScript, SQLite, Git

Data	Flood Prediction Dataset	Historical flood dataset with environmental parameters
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